Probability and Statistics
Topic 6. Continuous Random Variables

- Continuous random variable
- Probability density function
- Expectation of a continuous random variable
- Moment generating functions and their properties
- Uniform distribution
- Normal (Gaussian) distribution
- Exponential distribution
- Joint and marginal cumulative distribution function
- Joint and marginal probability density function
- Independent random variables
- Distribution of transformed random variables

Seminar 6

Problem 1. Assume that Z is a standard normal **r.v.** Find the **p.d.f.** of the following **r.v.**'s:

(a) $-Z$ (b) $2Z-3$; (c) $3-2Z$; (d) $\frac{Z-1}{4}$. (e) $|Z|$; (f) Z^2.

Problem 2. Assume that $X_1, X_2, \ldots, X_n$ are independent **r.v.**'s with distribution $\mathscr{U}(0,a)$. Find distribution (**c.d.f.** and **p.d.f.**) of the **r.v.**'s

(a) $U \sim \max\{X_1, \ldots, X_n\}$; (b) $V \sim \min\{X_1, \ldots, X_n\}$

Problem 3. Suppose that X and Y have joint probability density function

$$f_{X,Y}(x,y) = \begin{cases} 9e^{-3x-3y} & x \geq 0, y \geq 0; \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the marginal **p.d.f.**'s of X and Y; prove that X and Y are independent.

(b) find the joint distribution of the **r.v.**'s $S := X + Y$ and Y;

(c) find the distribution of S

(d) find the joint distribution of the **r.v.**'s $D := X - Y$ and Y;

(e) find the distribution of D.

Hint: in (b) and (d), use the formula for the **p.d.f.** of a transformed random vector $(X \pm Y, Y)$

Problem 4. Calculate the *moment generating function* (**m.g.f.**) $g_X(t) := \mathsf{E}(e^{tX})$ of a **r.v.** X with the

(a) uniform distribution $\mathscr{U}(a, b)$;

(b) exponential distribution $\mathscr{E}(\lambda)$;

(c) normal distribution $\mathscr{N}(\mu, \sigma^2)$

Problem 5. Alice and Bob enter simultaneously the grocery shop. Assume that their shopping times are independent **r.v.**'s X and Y respectively with exponential distribution of mean value 15 minutes.

(a) What is the distribution of the **r.v.** $M := \min(X, Y)$ for the smaller of the two shopping times? What is the expected value for the smaller shopping time?

(b) What is the distribution of the time $W := |X - Y|$ that the first to finish shopping will wait for his/her friend?

(c) Check that $\mathsf{E}(M) + \mathsf{E}(W)$ is larger than 15 min; is not there any contradiction there?

(d) Justify the strong memorylessness property: given that Alice is the second to finish her shopping, the difference $X - Y$ has the same distribution as X.

Problem 6. The joint probability density function of random variables X and Y is given by the formula

$$f_{X,Y}(x, y) = \begin{cases} a, & x, y > 0, x + y < 1 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of a

(b) Calculate the marginal **p.d.f.**'s f_X and f_Y and decide whether X and Y are independent

(c) Find $\mathsf{E}(X)$, $\mathsf{E}(Y)$ and the covariance $\mathsf{Cov}(X, Y)$

(d) Calculate $\mathsf{E}(XY)$

(e) Calculate the probability density function $f_{X|Y}(x \mid 0.5)$ of X given that $Y = 0.5$

(f) Find the conditional expectation $\mathsf{E}(X \mid Y = 0.5)$

Problem 7. Let $Z \sim N(0, 1)$, and let X be a random variable independent of Z with $P(X = 1) = P(X = -1) = \frac{1}{2}$. Define $Y = XZ$.

(a) Show that $Y \sim N(0, 1)$. *Hint: Condition on the value of X.*

(b) Show that Z and Y are dependent. *Hint: Compare $P(|Z| < 1,\ |Y| > 1)$ with $P(|Z| < 1)\, P(|Y| > 1)$. Note that $|Y| = |Z|$ always.*

(c) Show that Z and Y are uncorrelated, i.e. $\mathrm{Cov}(Z, Y) = 0$. *Hint:* $\mathrm{Cov}(Z, Y) = E[XZ^2]$. *Use the independence of Z and X.*

(d) Find $P(Z + Y = 0)$ and conclude that $Z + Y$ is not normally distributed, even though Z and Y are each normal. *Hint: Write $Z + Y = (1 + X)Y$ and consider the two cases $X = \pm 1$.*

Extra problems

Problem 8. Random variables X and Y are distributed according to the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} ax, & \text{if } 1 \le x \le 2 \text{ and } 0 \le y \le x, \\ 0, & \text{otherwise} \end{cases}$$

(a) Evaluate the constant a.

(b) Determine the marginal PDF $f_Y(y)$.

(c) Find $\mathsf{E}(X)$, $\mathsf{E}(Y)$ and the covariance $\mathsf{Cov}(X,Y)$

(d) Calculate $\mathsf{E}(XY)$

(e) Determine the conditional expectation of X given that $Y = 3/2$.

(f) Random variable Z is defined by $Z = Y - X$. Determine the PDF $f_Z(z)$

Problem 9. The metro train arrives at the station near your home every quarter hour starting at 6:00 a.m. You walk into the station every morning between 7:10 and 7:30 a.m., with the time in this interval being a random variable with given PDF. Let X be the elapsed time, in minutes, between 7:10 and the time of your arrival. Let Y be the time that you have to wait until you board a train. Calculate the **c.d.f.** of Y in terms of the **c.d.f.** of X and differentiate to obtain a formula for the **p.d.f.** of Y.

Problem 10. Romeo and Juliet have a date at a given time, and each, independently, will be late by an amount of time that is exponentially distributed with parameter α. What is the PDF of the difference between their times of arrival?